QUESTION ANALYSIS

EEE443

Power Plant Engineering

Question Analysis

Sessions 2020–21 *1 papers, 24 questions, 31 formulas*

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1. At a glance:

This analysis covers **1 paper** (2020-21), with 24 questions and 150 marks offered. It shows what past papers asked. It does not predict the next paper, and with so few papers a pattern can easily be chance.

With one paper, no trend can be read. Use this as a map of the topics.

The five topics with the most marks:

Topic	Marks	Share	Questions	Top subtopics (marks)
Steam power plant	30	20 %	5	Boiler efficiency (8); Layout (7); Improvements (5)
Load curves and factors	29	19 %	5	Factors (15); Base and peak load (10); Variable load (4)
Hydro plants	20	13 %	3	Turbines (8); Civil works (6); Merits (6)
Economic load sharing	18	12 %	2	Incremental rates (10); Two units (8)
Tariffs	14	9 %	2	Two-part (9); Flat and meter rates (5)

In the topic pages, a formula name in bold is a key formula, the number after a subtopic is its marks offered, and a blue symbol in an equation links to its meaning in the symbol list, where the arrow leads back to the equation.

2. Heat map:

Marks offered, by topic and paper, in book order. A darker cell means more marks. Click a topic to open its page.

Topic and subtopic	2020-21	All
Load curves and factors	**29**	**29**
Factors	**15**	15
Base and peak load	**10**	10
Variable load	**4**	4
Economics of generation	**12**	**12**
Depreciation	**7**	7
Cost per kWh	**5**	5
Tariffs	**14**	**14**
Two-part	**9**	9
Flat and meter rates	**5**	5
Economic load sharing	**18**	**18**
Incremental rates	**10**	10
Two units	**8**	8
Fuels and handling	**11**	**11**
Pulverisation	**6**	6
Characteristics	**5**	5
Steam power plant	**30**	**30**
Boiler efficiency	**8**	8
Layout	**7**	7
Improvements	**5**	5
Boilers	**5**	5
Superheaters and economisers	**5**	5
Steam turbines	**6**	**6**
Impulse and reaction	**6**	6
Combined cycle plants	**5**	**5**
Operation	**5**	5

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Topic and subtopic	2020-21	All
Hydro plants	**20**	**20**
Turbines	**8**	8
Civil works	**6**	6
Merits	**6**	6
Nuclear plants	**5**	**5**
Reactors	**5**	5

3. Priority list:

Must know: Steam power plant (30), Load curves and factors (29), Hydro plants (20), Economic load sharing (18), Tariffs (14), Economics of generation (12), Fuels and handling (11), Steam turbines (6), Combined cycle plants (5), Nuclear plants (5)

Likely: none

Less often: none

The rule: Must know means asked in every paper. Likely means asked in at least half the papers. Less often means asked in fewer than half. Numbers in brackets are marks offered, and each list runs from most marks to fewest.

4. Topic pages:

4.1. Load curves and factors:

29 marks, 5 questions, in 1 of 1 papers. Book: Mehta Ch. 3.

Most often theory (4 of 5). Subtopics: Base and peak load; Factors; Variable load.

Formulas:

Subtopic	Formula	Equation	Used in
Base and peak load (10)	**Average load**	$P_{av} = \frac{E}{t}$	20-21 1(d)
	Load factor	$\text{LF} = \frac{P_{av}}{P_{max}}$	20-21 1(d)
	Diversity factor	$\text{DF} = \frac{\sum P_{max,i}}{P_{max}} \geq 1$	20-21 1(d)
	Demand factor	$\frac{P_{max}}{\text{connected load}} \leq 1$	20-21 1(d)
Factors (15)	**Capacity factor**	$\text{CF} = \frac{P_{av}}{C} = \frac{E}{C\,t}$	20-21 4(c)
	Plant use factor	$\text{UF} = \frac{E}{C \times t_{op}}$	20-21 4(a)
	Utility (utilisation) factor	$\frac{P_{max}}{C}$	20-21 4(a)
	Reserve capacity	$C - P_{max}$	20-21 4(c)
	Energy at full load as per schedule	$E_{sched} = C\,t_{op} = \frac{E}{\text{UF}}$	20-21 4(c)

Symbols:

E = energy in the period (kWh) ↩

t = hours in the period (8760 for a year, 24 for a day) ↩

P_{max} = maximum demand on the station ↩

$P_{max,i}$ = maximum demand of consumer group i ↩

C = plant capacity (rated) ↩

t_{op} = hours the plant was in operation ↩

Questions:

No.	Paper	Marks	Type	Question
Base and peak load				
1	20-21 1(d)	10	theory	What are the requirements of a base load and peak load power plant? The annual peak load on a 30 MW power station is 25 MW. The power station supplies loads having a maximum demands of 10MW, 8.5 MW, 5 MW and 4.5 MW. The annual load factor is 45%. Find: (i) Average load, (ii) Energy supplied every year, (iii) Diversity factor, (iv) Demand/factor. *[sic: should be "Demand factor"]*
Factors				
2	20-21 2(a)	3	theory	What is the importance of load factor and diversity factor?
3	20-21 4(a)	4	theory	Define plant use factor and plant utility factor.
4	20-21 4(c)	8	numerical	A generating station has a maximum demand of 20 MW, a load factor of 60%, a plant capacity factor of 48% and a plant use factor of 80%. Find: (i) The daily energy produced (ii) The reserve capacity of the plant (iii) The maximum energy that could be produced daily if the plant was running all the time (iv) The maximum energy that could be produced daily if the plant was running fully loaded and operating as per schedule.
Variable load				
5	20-21 2(b)	4	theory	Why is the load on a power station variable? What are the effects of variable load on the operation of the power station?

4.2. Economics of generation:

12 marks, 2 questions, in 1 of 1 papers. Book: Mehta Ch. 4.

Most often numerical (1 of 2). Subtopics: Cost per kWh; Depreciation.

Formulas:

Subtopic	Formula	Equation	Used in
Cost per kWh (5)	Energy in a year	$E = C \times \text{LF} \times 8760$	20-21 5(b)
	Annual fixed charges	$(i + d)\,K + W$	20-21 5(b)
	Cost per kWh	$\dfrac{\text{annual fixed charges}}{E} + \text{running cost per kWh}$	20-21 5(b)
Depreciation (7)	Straight line	$D = \dfrac{P - S}{n}$ per year	20-21 6(a)
	Diminishing value	$P(1-r)^n = S \;\Rightarrow\; r = 1 - \left(\dfrac{S}{P}\right)^{1/n}$	20-21 6(a)
	Sinking fund	$q = (P - S)\,\dfrac{i}{(1+i)^n - 1}$	20-21 6(a)

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Symbols:

K = capital cost ↩

i, d = interest and depreciation rates ↩

W = wages, taxes ↩

P = initial cost ↩

S = salvage value ↩

n = useful life (years) ↩

r = fraction of the value left, charged each year ↩

q = annual deposit ↩

i = interest rate ↩

Questions:

No.	Paper	Marks	Type	Question
Cost per kWh				
1	20-21 5(b)	5	numerical	Estimate the generating cost per KWh delivered from a generating station from the following data: Plant capacity = 50 MW. Annual load factor = 40%. Capital cost = 1.2 crores. Annual cost of wages, taxation, etc. = Taka 4 lakhs. Cost of fuel, lubrication, maintenance, etc. = 1.0 paisa/kWh generated. Interest 5% per annum, depreciation 6% per annum of initial value.
Depreciation				
2	20-21 6(a)	7	theory	What are the methods of determining depreciation cost in power plants? Briefly explain one of the methods.

4.3. Tariffs:

14 marks, 2 questions, in 1 of 1 papers. Book: Mehta Ch. 5.

Most often theory (1 of 2). Subtopics: Flat and meter rates; Two-part.

Formulas:

Subtopic	Formula	Equation	Used in
Flat and meter rates (5)	Flat demand rate	$\text{bill} = a \times \text{kW}$	20-21 5(a)
	Straight line meter rate	$\text{bill} = b \times \text{kWh}$	20-21 5(a)
Two-part (9)	**Two-part tariff**	$\text{bill} = A \times \text{kW} + B \times \text{kWh}$	20-21 5(d)
	Demand charge	$A = \dfrac{\text{annual fixed charges}}{\text{aggregate maximum demand of consumers}}$	20-21 5(d)
	Energy charge	$B = \dfrac{\text{annual running charges}}{(1 - \text{loss})\, E_{gen}}$	20-21 5(d)

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No.	Paper	Marks	Type	Question

Symbols:

a = Tk per kW connected ↩ b = Tk per kWh ↩

Questions:

No.	Paper	Marks	Type	Question
Flat and meter rates				
1	20-21 5(a)	5	theory	Write short notes on the following type of tariffs: (i) Flat demand rate (ii) Straight line meter rate
Two-part				
2	20-21 5(d)	9	numerical	An electric supply company having a maximum load of 50 MW generates 18×10^7 units per annum and the supply consumers have an aggregate demand of 75 MW. The annual expenses including capital charges are: For fuel = Taka 90 lakhs. Fixed charges concerning generation = Taka 28 lakhs. Fixed charges concerning transmission and distribution = Taka 32 lakhs. Assuming 90% of the fuel cost is essential to running charges and the loss in transmission and distribution as 15% of KWh generated, deduce a two part tariff to find the actual cost of supply to the consumers.

4.4. Economic load sharing:

18 marks, 2 questions, in 1 of 1 papers. Book: Nag Ch. 1; Mehta Ch. 3.

Most often derivation (1 of 2). Subtopics: Incremental rates; Two units.

Formulas:

Subtopic	Formula	Equation	Used in
Incremental rates (10)	Heat rate	$\frac{I}{P}$	20-21 2(c)
	Incremental heat rate	$\frac{dI}{dP}$; in money $\frac{dF}{dP}$ (Tk/MWh)	20-21 2(c)
	Economic sharing	$\frac{dI_1}{dP_1} = \frac{dI_2}{dP_2} = \cdots = \lambda,\ \sum P_i = P$	20-21 2(c)
Two units (8)	Two linear units	$P_A = \frac{a_B P + b_B - b_A}{a_A + a_B}$	20-21 2(d)
	Cost of moving load	$\Delta F = \int_{P_1}^{P_2} \frac{dF}{dP}\, dP = \frac{a}{2}(P_2^2 - P_1^2) + b(P_2 - P_1)$	20-21 2(d)

Symbols:

I = heat input (kcal/h) ↩ F = fuel cost (Tk/h) ↩

P = output (MW) ↩ λ = common incremental rate ↩

Questions:

No.	Paper	Marks	Type	Question
Incremental rates				
1	20-21 2(c)	10	derivation	Show that for the economic load sharing, the incremental heat rates are equal.

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No.	Paper	Marks	Type	Question
Two units				
2	20-21 2(d)	8	numerical	The incremental fuel costs for two generating units A and B of a power plant are given by the following equations: $\frac{dF_A}{dP_A} = 0.06P_A + 11.4$, $\frac{dF_B}{dP_B} = 0.07P_B + 10$, where P is in mega-watts and F is in takas per hour. (i) Find the economic loading of the two units when the total load to be supplied by the power station is 150 MW. (ii) Find the loss in fuel cost per hour if the load is equally shared by the two units.

4.5. Fuels and handling:

11 marks, 2 questions, in 1 of 1 papers. Book: Nag Ch. 4, 5.

Most often theory (2 of 2). Subtopics: Characteristics; Pulverisation.

Questions:

No.	Paper	Marks	Type	Question
Characteristics				
1	20-21 1(b)	5	theory	Briefly discuss the characteristics of fuels.
Pulverisation				
2	20-21 6(b)	6	theory	Why is pulverization done in a coal fired power plant? Give description of any basic type of pulverizers.

4.6. Steam power plant:

30 marks, 5 questions, in 1 of 1 papers. Book: Nag Ch. 6, 8.

Most often theory (4 of 5). Subtopics: Improvements; Layout; Boilers; Superheaters and economisers; Boiler efficiency.

Formulas:

Subtopic	Formula	Equation	Used in
Boiler efficiency (8)	Heat from the coal	$Q_{in} = m_f\, C$	20-21 3(d)
	Boiler efficiency	$\eta_{boiler} = \frac{m_w(h - h_f)}{m_f\, C}$	20-21 3(d)
	Economiser efficiency	$\eta_{eco} = \frac{m_w c_w(t_2 - t_1)}{m_g c_{pg}(t_g - t_a)}$	20-21 3(d)
	Whole boiler plant	$\eta_{plant} = \frac{m_w(h - c_w t_1)}{m_f\, C}$	20-21 3(d)
	Gas temperature drop	$\Delta t_g = \frac{m_w c_w(t_2 - t_1)}{m_g c_{pg}}$	20-21 3(d)

Symbols:

m_f = coal burnt (kg/h) ↩

C = calorific value (kcal/kg) ↩

m_w = steam (water) per hour (kg/h) ↩

h = enthalpy of the steam ↩

h_f = enthalpy of the water entering the boiler (kcal/kg) ↩

t_1, t_2 = water in and out ↩

m_g, c_{pg} = flue-gas mass and specific heat ↩

t_g = gas entering ↩

t_a = atmosphere ↩

Questions:

No.	Paper	Marks	Type	Question
Improvements				
1	20-21 1(a)	5	theory	List the factors that contribute to improving the generating capacity of a thermal power station.
Layout				
2	20-21 3(a)	7	theory	Draw the schematics of steam generating part of a coal fire power plant showing all basic components and direction of steam flow.
Boilers				
3	20-21 3(b)	5	theory	Differentiate between water tube boiler and fire tube boiler.
Superheaters and economisers				
4	20-21 3(c)	5	theory	A superheater and an economizer improve the performance of a boiler-How?
Boiler efficiency				
5	20-21 3(d)	8	numerical	The following data is supplied for a boiler plant: Boiler: Mass of coal supplied = 220 kg /hr. Calorific value of coal = 7200 kcal /kg. Mass of feed water = 200 kg/hr *[sic: should be 2000 kg/hr]*. Enthalpy of steam produced = 670 kcal/kg. Economizer: Inlet temperature of feed water = 18°C. Outlet temperature of water = 88°C. Atmospheric temperature = 19°C. Temperature of flue gas entering = 350°C. Mass of flue gases = 4150 kg/hr. Determine the following (i) Efficiency of boiler. (ii) Efficiency of economizer. (iii) Efficiency of whole boiler plant.

4.7. Steam turbines:

6 marks, 1 question, in 1 of 1 papers. Book: Nag Ch. 7.

Most often theory (1 of 1). Subtopics: Impulse and reaction.

Questions:

No.	Paper	Marks	Type	Question
Impulse and reaction				
1	20-21 6(c)	6	theory	Write short notes on: (i) Impulse turbine (ii) Reaction turbines

4.8. Combined cycle plants:

5 marks, 1 question, in 1 of 1 papers. Book: Nag Ch. 3.

Most often theory (1 of 1). Subtopics: Operation.

Questions:

No.	Paper	Marks	Type	Question
Operation				
1	20-21 1(c)	5	theory	What is meant by combined cycle power generation?

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4.9. Hydro plants:

20 marks, 3 questions, in 1 of 1 papers. Book: Nag Ch. 10; Mehta Ch. 2.

Most often theory (3 of 3). Subtopics: Turbines; Civil works; Merits.

Formulas:

Subtopic	Formula	Equation	Used in
Turbines (8)	**Water power**	$P = \rho g Q H$ (W); output $= \eta\, \rho g Q H$	20-21 4(d)

Symbols:

$\rho = 1000$ kg/m^3 ↩ $Q =$ flow (m^3/s) ↩ $\eta =$ overall efficiency ↩

$g = 9.81$ m/s^2 ↩ $H =$ head (m) ↩

Questions:

No.	Paper	Marks	Type	Question
Turbines				
1	20-21 4(d)	8	theory	With diagram, write short notes on Kaplan and Francis turbine.
Civil works				
2	20-21 5(c)	6	theory	Write short note on- (i) Spillways (ii) Surge tanks (iii) Draft tubes
Merits				
3	20-21 6(d)	6	theory	Discuss the advantages and disadvantages of a Hydro-electric power plant.

4.10. Nuclear plants:

5 marks, 1 question, in 1 of 1 papers. Book: Nag Ch. 9; Mehta Ch. 2.

Most often theory (1 of 1). Subtopics: Reactors.

Questions:

No.	Paper	Marks	Type	Question
Reactors				
1	20-21 4(b)	5	theory	Mention the types of reactors used in nuclear power plants.

5. Derivations:

Every derivation asked, in book order. Each outline gives the steps; the link opens the full working.

No.	Paper	Marks	Derivation	Outline
Economic load sharing				
1	20-21 2(c)	10	Incremental rates	Fix the total load; set the change of total heat input to zero; show the slopes are equal.

6. Repeat tracker:

No repeats among these papers.

7. Question style:

How each topic was asked, as a number of questions.

Topic	Theory	Derivation	Numerical
Load curves and factors	4	·	1
Economics of generation	1	·	1
Tariffs	1	·	1
Economic load sharing	·	1	1
Fuels and handling	2	·	·
Steam power plant	4	·	1
Steam turbines	1	·	·
Combined cycle plants	1	·	·
Hydro plants	3	·	·
Nuclear plants	1	·	·

8. Study order:

Topics by marks offered, with the share of all marks covered so far. Papers offer more marks than you answer, so this is a guide to where marks have been, not a score.

No.	Topic	Marks	Covered so far
1	Steam power plant	30	20 %
2	Load curves and factors	29	39 %
3	Hydro plants	20	53 %
4	Economic load sharing	18	65 %
5	Tariffs	14	74 %
6	Economics of generation	12	82 %
7	Fuels and handling	11	89 %
8	Steam turbines	6	93 %
9	Combined cycle plants	5	97 %
10	Nuclear plants	5	100 %

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